LAW-E

Thermodynamic Governance Framework

Executive Brief

Runtime AI Stability — From Observation to Governance

The Problem

> *Every AI system in production carries a disclaimer. That disclaimer is not a legal formality — it is an admission: there is no way to know, in real time, whether the system is stable or not.*

Most organizations detect AI failures after they occur. At that point, the cost is already incurred.

For organizations deploying AI in production, this creates three unresolved challenges:

⚠ **Operational risk**
Hallucinations, cascading errors, and reasoning failures are undetectable until they cause damage.

💰 **Cost unpredictability**
Without runtime visibility, computational waste and failure remediation costs cannot be anticipated or controlled.

⚖ **Regulatory exposure**
The EU AI Act requires continuous monitoring, traceability, and auditability for high-risk AI systems — with no existing standard instrument to deliver this.

The Solution — Law-E

Law-E is a runtime thermodynamic governance framework defining a new class of measurable stability signals for AI systems.

Instead of assuming stability, it measures it — continuously, at inference time, without modifying the underlying model.

This results in a unified governance variable integrating instantaneous state, accumulated instability, and runtime conditions.

Observe	Measure	Govern
Runtime instability signals captured and normalized	A unified stability indicator (ESI) computed in real time	Authorize, modulate, or block based on stability

How It Works

This pipeline transforms raw execution signals into actionable governance decisions.

> *Runtime Signals → Normalizer → Thermodynamic Flow → Aggregation → Governance Decision*

Two operational modes:

OBS Mode — Monitoring only — measure, log, and audit system stability without intervention

GOV Mode — Active governance — regulate system behavior in real time based on stability indicators

Experimental Results

TruthfulQA benchmark — ~790 questions	
Traditional systems	✗ No runtime instability signal ✗ Failures detected after impact
~79% Law-E	✓ Instability signals detected at runtime — before impact

EU AI Act Alignment

> **Law-E directly addresses EU AI Act requirements:**
>
> ✓ Continuous performance monitoring of deployed systems

- ✓ Traceability and structured audit logs
- ✓ Runtime risk assessment for high-risk AI applications
- ✓ Post-deployment monitoring and reporting

Intellectual Property

Law-E is protected by a structured patent portfolio filed with INPI (France), covering five complementary invention classes:

- Thermodynamic measurement systems
- Temporal aggregation mechanisms (memory)
- Stability-constrained training architectures (learning)
- Distributed thermodynamic coordination (network)
- Unified governance layer (decision)

Engage with Neomundi Lab

We are onboarding a limited number of early partners to:

- ✓ Measure AI stability in production environments
- ✓ Evaluate governance impact on real workflows
- ✓ Prepare for compliance-ready AI deployment

OBS pilot	Integrate runtime stability monitoring into your existing AI infrastructure — zero model modification required
GOV pilot	Deploy active governance on a controlled AI workflow — measure the impact on reliability and compliance posture
Research collaboration	Explore thermodynamic stability in your domain — LLM infrastructure, multi-agent systems, distributed environments

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Reliability is no longer assumed. It is continuously evaluated.

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One stability law, many domains of application